Minjae Lee

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Research Interests

- Geometrically-Consistent, Video-Based World Models
- 2D/3D/4D Computer Vision, LiDAR-Based Autonomous Driving, SLAM

Experience

- **ETRI Visual Intelligence Lab** — Daejeon, South Korea
  Post-Graduate Researcher — Aug. 2026 – Present
  - Video-Based World Model Training & Embodied AI Action Model Development

Education

- **Gyeongsang National University, Graduate School** — Jinju, South Korea
  M.S. in Computer Science and Engineering (GPA: 4.50/4.50) — Mar. 2025 – Aug. 2026
- **Gyeongsang National University** — Jinju, South Korea
  B.S. in Computer Science (GPA: 4.32/4.50) — Mar. 2020 – Feb. 2025

Languages

- **Korean**: Native
- **English**: TOEIC Speaking Intermediate High / Conversational and capable of Q&A at international conferences

Projects

- **LiDAR-Based Autonomous Driving Perception**
  - LoRa-MOS: Density-Decay-Robust LiDAR Moving Object Segmentation (IEEE RA-L) — Feb. 2026 – Sep. 2026
  - X-MOS: Cross-LiDAR Generalization Framework (ICRA 2026) — Aug. 2025 – Jan. 2026
  - SwiftMOS: Real-Time LiDAR Moving Object Segmentation (IEEE RA-L) — Jan. 2025 – Jan. 2026
- **Camera-Based Computer Vision Applications**
  - Finger-Pointing-based Small Robot Control (SIGGRAPH Asia 2024) — Mar. 2024 – Dec. 2024
  - Shrunken Reality: SLAM-based Real-Time Real-World Context Augmentation (SIGGRAPH Asia) — Jan. 2024 – Dec. 2024
  - SLAM-based Illegal Parking Detection System (SIGGRAPH Asia 2024) — Aug. 2023 – Dec. 2024
  - Low-Cost Augmented Reality Interaction (ETRI) — Jan. 2023 – Dec. 2023

Honors & Awards

- **Manufacturing AI Competition** – Factory Equipment Noise Anomaly Detection (**Excellence Award**) — May 2026
- **Gyeongsangnam-do Governor's Award** – IT Engineering College Commencement (**Valedictorian**) — Feb. 2025
- **Naver 1784 School Revolution** – Elementary AI Education, National 1st Place (**Grand Prize**) — Nov. 2023
- **Likelion Hackathon** – Web Development, National 3rd Place (**Bronze**) — Aug. 2023

Publications

First author: 10, Co-author: 12

[International Journals]

- **Lee, M.**, et al. (2026). LoRa-MOS: Density-Decay-Robust LiDAR Moving Object Segmentation via Logarithmic Spatial Representation and Range-Balanced Learning. *IEEE Robotics and Automation Letters*.
- Kim, U., **Lee, M.**, & Lee, S. (2026). TT-FSI: Scalable Faithful Shapley Interactions via Tensor-Train. *IEEE Access*.

- Bae, J., **Lee, M.**, Buu, S. J., Choi, S. M., & Lee, S. (2026). Enhancing Monocular 3D Semantic Scene Graph Prediction via Cross-Modality Knowledge Distillation. *Pattern Recognition*.
- Park, S., **Lee, M.**, Kim, U., Bae, J., & Lee, S. (2026). Early Detection of Failure Onset in 3D Printing: A Foundation-Segmentation Pipeline. *Engineering Applications of Artificial Intelligence*.
- **Lee, M.**, Kim, U., Kim, G. W., & Lee, S. (2026). SwiftMOS: A Fast and Lightweight Moving Object Segmentation via Feature Flowing Direct View Transformation. *IEEE Robotics and Automation Letters*.
- **Lee, M.**, Lee, S., & Seo, H. (2026). Enhancing Early Alzheimer's Disease Detection via Transfer Learning: From Big Structural MRI Datasets to Ethnically Distinct Small Cohorts. *Applied Sciences*, 16(2), 1004.
- Yeon, S., **Lee, M.**, Bae, J., & Lee, S. (2026). Resolving Ambiguity in Pointing Gestures Using Contextual Reasoning from Large Language Models. *Computer Modeling in Engineering & Sciences*.

[**International Conferences**]

- **Lee, M.**, Ha, I., Choi, S. M., Kim, G. W., & Lee, S. X-MOS: A Heterogeneous Cross-LiDAR Generalization Framework for Moving Object Segmentation. *IEEE International Conference on Robotics and Automation (ICRA) 2026*.
- **Lee, M.**, Choi, S. M., Kim, G. W., & Lee, S. HBRB-BoW: A Retrained Bag-of-Words Vocabulary for ORB-SLAM via Hierarchical BRB-KMeans. *IEEE International Conference on Robotics and Automation (ICRA) 2026* (Late-Breaking Results).
- Kang, W., Kim, M., **Lee, M.**, Kim, U., Bae, J., Lee, S., Kim, G. W., & Choi, S. M. Hierarchical Interests Modeling on Multiple Time Scales for Sequential Recommendation. *Artificial Intelligence with Biased or Scarce Data Workshop at AAAI 2026*.
- Kim, U., **Lee, M.**, Bae, J., Kim, M., Choi, S. M., & Lee, S. Privacy-Preserving Argumentative Explanations. *AAAI 2026*.
- **Lee, M.**, Bae, J., Kim, U., Choi, S. M., & Lee, S. Shrunken Reality: Augmenting Real-World Contexts in Real-Time on Realistic Miniature Dioramas. *SIGGRAPH Asia 2024*.
- Bae, J., **Lee, M.**, Kim, U., & Lee, S. SLAM-Based Illegal Parking Detection System. *SIGGRAPH Asia 2024*.
- **Lee, M.**, Bae, J., Choi, S. M., & Lee, S. Finger-Pointing Interface for Human Gesture Recognition Based on Real-Time Geometric Comprehension. *SIGGRAPH Asia 2024*.

[**Domestic Journals**]

- **Lee, M.**, Bae, J., & Lee, S. (2025). Watch My Wrist: Wrist Muscle Movement-based Hand Gesture Recognition for Out-of-Sight in Augmented Reality. *The Journal of the Korean Institute of Information Technology*, 23(7), 167–170. (in Korean)
- **Lee, M.**, Bae, J., Choi, S. M., & Lee, S. (2024). Plane Detection and Projection-based Filter for Enhanced Moving Object Segmentation in 3D LiDAR Environments. *The Journal of the Korean Institute of Information Technology*, 22(12), 19–27. (in Korean)

[**Domestic Conferences**]

- Ha, I., **Lee, M.**, & Lee, S. Image-Processing-based Hand-Tremor Compensation Pointing System for Digital Presentation Environments. *KIIT Fall Conference 2025*. (in Korean)
- Yeon, S., **Lee, M.**, & Lee, S. Low-Cost Mixed Reality (MR) System based on 3D Hand Tracking. *KIIT Fall Conference 2024*. (in Korean)
- Kim, U., **Lee, M.**, & Lee, S. Multiplayer System Combining First-Person Mixed Reality and Real-Time Strategy. *KIIT Fall Conference 2024*. (in Korean)
- **Lee, M.**, Bae, J., & Lee, S. Plane-Detection-Filter-based Moving Object Segmentation in 3D LiDAR Environments. *KIIT Fall Conference 2024*. (in Korean)
- Kim, U., Bae, J., **Lee, M.**, & Lee, S. Explainable Facial Beauty Prediction Model. *KIIT Fall Conference 2023*. (in Korean)
- Bae, J., **Lee, M.**, Kim, U., & Lee, S. SLAM-based Illegal Parking Detection System. *KIIT Fall Conference 2023*. (in Korean)